

Quanyan Zhu

Room 1004, 370 Jay St., Brooklyn, NY Email: qz494@nyu.edu Web: wp.nyu.edu/quanyan
ScholarGPS: scholargps.com/scholars/54961215509101/quanyan-zhu
Research.com: research.com/u/quanyan-zhu
Google Scholar: scholar.google.com/citations?user=Qt2WDh0AAAAJ

Employment

- Associate Professor, Department of Electrical and Computer Engineering, New York University, NY, USA, Sept. 2019–Present
- Affiliated Associate Professor, Center for Urban Science And Progress (CUSP), New York University, NY, USA, Sept. 2019–Present
- Assistant Professor, Department of Electrical and Computer Engineering, New York University, NY, USA, Jan. 2014–Aug. 2019
- Affiliated Assistant Professor, Center for Urban Science And Progress (CUSP), New York University, NY, USA, April 2019–Aug. 2019
- Postdoctoral Research Associate, Department of Electrical Engineering, Princeton University, NJ, USA, Sept. 2013–Dec. 2013 (Host: Vincent Poor)

Research Impact and Scholarly Metrics

- ScholarGPS® ID: 54961215509101. Highly Ranked Scholar – Lifetime: #11 in Game Theory and #12 in Physical Systems; Highly Ranked Scholar – Prior Five Years: #2 in Game Theory.
- Research.com Computer Science rankings: Current World Ranking #4399 and Current National Ranking #2046.
- Google Scholar citation metrics: 17,466 citations (10,149 since 2021); h-index 65 (48 since 2021); i10-index 283 (233 since 2021). Retrieved August 10, 2026.
- Internationally recognized researcher in game theory, cybersecurity, cyber-physical systems, AI security, resilient autonomy, and network science.
- Research contributions span strategic learning, control theory, cyber deception, federated learning, resilient AI, and multi-agent systems; developed foundational cyber-resilience frameworks integrating game theory, control theory, and learning theory.
- Known for influential work on defensive deception, resilient control, moving target defense, and AI-enabled cyber resilience.

Research Interests

- Dynamic Games and Control
- Multi-Agent Systems and Reinforcement Learning
- Human-Machine Interactions
- Cyber Resiliency
- Distributed Decision-Making and Mechanism Design
- Cyber-Physical-Human Systems
- Critical Infrastructure Protection
- Internet of Things

- Cyber Manufacturing
- Security of 5G Networks
- STEM Education

Education

- Doctor of Philosophy, Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign, USA, Aug. 2008–May 2013. Ph.D. Dissertation: Game-Theoretic Methods for Security and Resilience of Cyber-Physical Systems, advised by Prof. Tamer Başar.
- Master of Applied Science in Electrical Engineering, University of Toronto, ON, Canada, Sept. 2006–June 2008. Master Thesis: An Investigation on Link Level End-to-End Power Control Models in WDM Optical Networks, advised by Prof. Lăcră Pavel.
- Bachelor of Engineering in Honors Electrical Engineering, McGill University, Montréal, QC, Canada, Sept. 2002–June 2006.

Current and Past Students

Academic Placements as Assistant Professors

- Ya-Ting Yang — Assistant Professor, National Taiwan University.
- Tao Li — Assistant Professor, City University of Hong Kong.
- Shutian Liu — Assistant Professor, City University of Hong Kong.
- Linan Huang — Assistant Professor, Tsinghua University.
- Juntao Chen — Assistant Professor, Fordham University.
- Junaid Farooq — Assistant Professor, University of Michigan.

Current PhD Students

- Ya-Ting Yang, Ph.D. Candidate; Assistant Professor, National Taiwan University.
- Yunian Pan, Ph.D. Candidate
- Zhengye Han, Ph.D. Student

Visiting Students

- Anh Tung Nguyen, Uppsala University, Sweden.

Postdoctoral Researcher

- Song Fang (Oct. 2019–Aug. 2021)

Past Visiting Students

- Yeongwoo Kim, Royal Institute of Technology (KTH), Sweden (Fall 2023)
- Kim Hammar, Royal Institute of Technology (KTH), Sweden (May 2023)
- Liuwei Zhou, Ph.D. Student, Donghua University, Shanghai, China (2015–2016)
- Thi Thu Hang Nguyen, Ph.D. Student, LAAS-CNRS, France (Summer 2017)

Graduated PhD Students

- Tao Li, Ph.D. (2024). Assistant Professor, City University of Hong Kong. Winner of the 2024 Dante Youla Award for Graduate Research Excellence.
- Shutian Liu, Ph.D. (2024). Assistant Professor, City University of Hong Kong. Dissertation: System-Scientific Foundations of Holonic Risk Analysis and Design in Socio-Cyber-Physical Systems.
- Yuhan Zhao, Ph.D. (2024). Dissertation: Learning and Game Theoretic Paradigms for Strategic Coordination in Multi-Agent Autonomous Systems.
- Yinan Hu, Ph.D. (2023). Dissertation: Advanced and Contemporary Applications of Classical and Quantum Detection Theories in Strategic Environments.
- Yunfei Ge, Ph.D. (2024). Dissertation: The Symbiosis of Trust and AI. Winner of the 2021 Chang Education Award; Winner of the 2024 Goodman Research Award.
- Tao Zhang, Ph.D. (2023). Dissertation: Dynamic Mechanism Design: From Theories to Applications. Winner of the 2020 Goodman Research Award.
- Linan Huang, Ph.D. (2022). Assistant Professor, Tsinghua University. Dissertation: Strategic Socio-Technical Defense Mechanisms for High-Confidence Cyber-Physical Systems. Winner of the 2022 Youla Award; Winner of the 2023 NYU-Wide Dissertation Award.
- Yunhan Huang, Ph.D. (2022). Dissertation: Sequential Decision Making with Strategic Information Acquisition and Adversarial Manipulation.
- Juntao Chen, Ph.D. (2020). Assistant Professor, Fordham University. Dissertation: Trustworthy Cyber-Physical Networks by Design. Winner of the 2019 Youla Award; Winner of the 2021 Hessel Dissertation Award.
- Junaid Farooq, Ph.D. (2020). Assistant Professor, University of Michigan. Dissertation: Cyber-Physical Dynamic Decision Mechanisms for Large-Scale IoT Systems. Winner of the 2019 Papoulis Award; 2020 Youla Award; 2021 NYU Dissertation Award.
- Rui Zhang, Ph.D. (2020). Dissertation: Game-Theoretic Approaches to Cyber Data Risk Mitigation and Transfer. Winner of the 2016 Rosenthal Award.
- Jeff Pawlick, Ph.D. (2018). Dissertation: A System Science Perspective on Deception for Cybersecurity in the Internet of Things.
- Zhiheng Xu, Ph.D. (2018). Dissertation: Security and Resilience of Cyber-Physical Systems.

Supervised MS Theses

- Pengyi Jiang, M.S. (2026). Thesis: Semantic Cooperative Game: Semantic Value Allocation and Coalition Formation in Multi-Agent LLM Collaboration, Master of Science in Electrical Engineering, New York University Tandon School of Engineering.
- Shizhen Jia, M.S. (2024). Thesis: Interference Avoidance for FMCW Automotive Radars Using Non-Cooperative Game Theory. Now Ph.D. Student at NYU.
- Lan Shi, M.S. (2022). Thesis: Heterogeneous Robots Collaborative Applications with Asymmetric Reinforcement Learning. Now Ph.D. Student at Purdue University.
- Chen Peng, M.S. (2021). Thesis: Interplay Between Trust and Networks: From Pandemics to 5G. Now Ph.D. Student at Purdue University.
- Alex Dunyak, M.S. (2019). Thesis: A Hamiltonian Approach to Stochastic Minimization. Winner of the 2020 Tamir Award.
- Guanze Peng, M.S. (2018). Thesis: Quickest Change Detection for False Data Injection Attacks. Winner of the 2019 Tamir Award.

Graduated MS Students

- Katherine Zhuo, Summer M.S. Student (2026). Research: Agentic AI risks and insurance.
- Xinhong Xie, M.S. (2024). Research: LLMs for Evading Censorship. Now Ph.D. Student at Penn State University.
- Haozhe Lei, M.S. (2022). Research: Black-Box Attacks in Meta Reinforcement Learning. Now Ph.D. Student at NYU.
- Zhili Zhang, M.S. (2021). Research: Games of Partial Information for Cyber Deception. Now Ph.D. Student at UConn.
- Timothy Kieras, M.S. (2020). Research: IoT Supply Chain Risks.
- Yunian Pan, M.S. (2020). Research: Meta-Learning for Multi-Player Games. Now Ph.D. Student at NYU.
- Alex Dunyak, M.S. (2019). Research: Stochastic Minimum Principle. Now Ph.D. Student at McGill University. Winner of the 2020 Tamir Award.
- Guanze Peng, M.S./Ph.D. Track (2018). Research: Quickest Change Detection in Linear Systems. Now Ph.D. Student at NYU.
- Zhuqing Liu, M.S. (2018). Research: Adversarial Learning for Multi-Agent Systems. Now Ph.D. Student at Iowa State University.
- Kai Zheng, M.S. (2018). Research: IoT-Enabled Fridge Implementation.
- Hamza Anwar, M.S. (2017). Research: Multi-Scale H_∞ Control and Filtering. Now Ph.D. Student at Ohio State University.
- Rui Zhang, M.S. (2016). Research: Adversarial Machine Learning. Winner of the Rosenthal Award. Now Ph.D. Student at NYU.

Past Undergraduate Students

- Minjun Kang (2025): Agentic AI for proactive cybersecurity
- Xuan Liu (2024): Prosocial Irrationality for LLM Agents.
- Haoyang Shang (2024): Cyber Operations Research Gym (CybORG).
- Junjie Chen (2024): LLMs for Network Security.
- Juan Guevara (2023): Competitive RL as Sequence Modeling.
- Josh Schrock (2022): Game Theory and DMD of Cognitive Systems.
- Prajjwal Bhattarai (2021): Fake News in Epidemics.
- Andrew Paul Mayer (2021): Brain-Machine Interface Control.
- Zijie Wu (2020): NYC Food Shortages During COVID-19.
- Nurpeiis Baimukan (2019): Steganographic Text via Generative Models.
- Aidan Collins: VIP – Internet of the Controlled Things.
- Jacky Yan: VIP Internet of the Controlled Things.
- Shashank Gupta (2018): Machine Scheduling with Precedence Constraints.
- Zhanghao Chen (2018): Online Fake Review Generation.
- Rundong Chen (2018): Search Theory for On-Demand Transportation.
- Jin Shang (2018): Real-Time Mechanism Design for Wireless IoT Sensors.
- Teddy Zeng (2017): Citibike Resource Allocation.
- Zi Yuan Huang (2017): Security Game Design.
- Virinchi Roy (2017): Dynamic Mode Decomposition for Networks.

- Weiyu Wang (2017): Human-in-the-Loop Machine Learning.
- Muhammad Shujaat Mirza (2016): IoT Security and Privacy.
- Siba Siddique (2016): FEW Nexus Network Analysis.
- Ali Abouelatta (2016): Smart Fridge Design.
- Ahmed Nabil (2016): Control of Cloud-Enabled Quadrotors.
- Wai San Gu (2015): Moving Target Defense for Infrastructure.
- Franky Chen (2015): Deception in Cloud Robotics.
- Vasily Rudchenko (2015): Deception in Cloud Robotics.
- John Pak, Jatinder Singh, Saboor Zahir (2015): Protection and Deception Board Game. Winner of the NYU Prototyping Fund Award.

Teaching

Regular Courses

- ECE-GY 9953 Advanced Projects
- ECE-GY 9900 Seminar in Electrical and Computer Engineering
- ECE-GY 6233 System Optimization Methods
- ECE-GY 8233 Optimal Control
- ECE-UY 3064 Feedback Control
- ECE-GY 5213 Introduction to System Engineering

Developed Courses

- ECE-GY 6263 Game Theory
- ECE-GY 9213 Special Topics: Game Theory
- ECE-GY 9223 Special Topics: Reinforcement Learning
- ECE-GY 9233 Special Topics: Control and Optimization of Cyber-Physical Human Networks
- ECE-GY 9253 Special Topics: System Foundations of Cyber-Physical System Security
- VIP-UY 2001 Vertically Integrated Projects I: Internet of Controlled Things, 2017-2020

Courses under planning

- Artificial Intelligence and Games (Undergraduate and Graduate Course)
- Systems Engineering for Complex Problem-Solving (Graduate Course)
- Resilience System Engineering and Science (Graduate Course)

Short Courses, Tutorials, and Workshops

1. Organizer, Human-Aware AI Agents for the Cyber Battlefield: From Human Models to Autonomous Defense, AAAI 2026 Summer Symposium Series, June 22–24, 2026, Dongguk University, Seoul, South Korea.
2. Organizer, Tutorial, 23rd IFAC World Congress (IFAC WC 2026), August 23–28, 2026, Busan, Republic of Korea.
3. General Co-Chair, Workshop on Resilience in Next Generation Network Management (NGResNet), in con-

junction with IEEE/IFIP Network Operations and Management Symposium (NOMS 2026), Rome, Italy, May 18–22, 2026.

4. General Co-Chair, First Workshop on Resilience in Next-Generation Networks (NGResNet 2025), in conjunction with IEEE International Conference on Computer Communications (INFOCOM 2025), London, United Kingdom, May 19, 2025.
5. Workshop Chair and Co-Organizer, 1st–4th IEEE CNS Cyber Resilience Workshop (CRW), held with the IEEE Conference on Communications and Network Security (IEEE CNS), 2022–2025.
6. Organized Multiple Invited Sessions on CPS Security and Resilience, 2023 IEEE 62nd Conference on Decision and Control (CDC), December 13–15, 2023, Singapore.
7. Organized Workshop on Control, Game, and Learning Theory for Security and Privacy, 2023 IEEE 62nd Conference on Decision and Control (CDC), December 12, 2023, Singapore.
8. Organized Minisymposium on Recent Advances in Dynamic Games and Control Theory and Their Connection to Data Science, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan, August 20–25, 2023.
9. Organized Open Invited Track for Modeling and Design of Secure and Resilient Control Systems, IFAC World Congress 2023, July 9–14, Yokohama, Japan.
10. Organized Workshop on Interdisciplinary and Convergent Research Foundations for Cyber Deception, August 2–3, 2023, New York, NY, USA.
11. Organized Invited Session on Dynamic Games and Networked Systems, 2022 IEEE 61st Conference on Decision and Control (CDC), December 6–9, 2022, Cancun, Mexico.
12. Organized Invited Session on Learning and AI Methods for CPS Security, 2022 IEEE 61st Conference on Decision and Control (CDC), December 6–9, 2022, Cancun, Mexico.
13. Delivered Tutorial on IoT Supply Chain Security Risk Assessment and Mitigation: Methodologies and Computational Tools, Military Communications Conference (MILCOM), November 28–December 2, 2022, National Capital Region, USA.
14. Delivered Tutorial on Cybersecurity in Robotics, 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2021), September 27–October 1, 2021, Prague, Czech Republic (Virtual).
15. Organized Panel on Supply Chain Security for Industrial Control Systems, Resilience Week, October 18–21, 2021 (Virtual).
16. Organized Invited Sessions on Game Theory for Cyber Deception, 2021 IN- FORMS Annual Meeting, October 24–27, 2021 (Virtual).
17. Organized Invited Sessions on Game Theory for Cyber Security, 2020 INFORMS Annual Meeting, November 8–11, 2020 (Virtual).
18. Delivered Tutorial on Understanding IoT Security Risks and Resilience: From Networks to Supply Chain, IEEE 6th World Forum on Internet of Things, New Orleans, Louisiana, USA (Virtual).
19. Organized Workshop on Learning and Security for Multi-Agent Systems, 59th Conference on Decision and Control (CDC), December 13, 2020 (Virtual).
20. Organized Workshop on Security and Privacy in Robotics, International Conference on Robotics and Automation (ICRA), May 31–June 4, 2020, Paris, France (Virtual).
21. Organized Workshop on Learning, Games, and Control for Security of Cyber- Physical Systems, 58th Conference on Decision and Control (CDC), December 11–13, 2019, Nice, France.
22. Delivered Tutorial on Resource Management for On-Demand Mission-Critical Internet of Things Applications, IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), November 11–14, 2019, Newark, NJ, USA.
23. Delivered Tutorial on Game Theory for Cyber Deception, Symposium and Bootcamp on the Science of Security (HotSoS), April 2–3, 2019, Nashville, Tennessee, USA.

24. Organized Workshop on Secure and Resilient Control Systems, 57th IEEE Conference on Decision and Control (CDC), December 17-19, 2018, Miami Beach, FL, USA.
25. Organized Workshop on Game Theory Meets Network Security, 25th ACM Conference on Computer and Communications Security (CCS), October 15-19, 2018, Toronto, Canada.
26. Organized Workshop on Game Theory for Cyber Deception, 9th Conference on Decision and Game Theory for Security (GameSec), October 29-31, 2018, Seattle, WA, USA.
27. Delivered Tutorial on Game-Theoretic Methods for Modeling and Defending Against Advanced Persistent Threats in Wireless Systems, 16th International Symposium on Modeling and Optimization in Mobile, Ad Hoc and Wireless Networks (WiOpt), May 7, 2018, Shanghai, China.
28. Organized Panel on “Transformation to a Resilient Grid: From Research Philosophy to Multidisciplinary Education,” 2016 IEEE Power & Energy Society General Meeting, July 17-21, 2016, Boston, MA, USA.
29. Organized Special Session on Game Theory for Large-Scale Systems and Complex Networks, 54th IEEE Conference on Decision and Control (CDC), December 12- 14, 2016, Las Vegas, NV, USA.
30. Delivered Tutorial on Game Theory for Cyber Security, 8th IEEE International Workshop on Information Forensics and Security (WIFS), December 5-7, 2016, Abu Dhabi, UAE.
31. Organized Workshop on Secure Cyber-Physical Systems, 54th IEEE Conference on Decision and Control (CDC), December 15-18, 2015, Osaka, Japan.
32. Delivered Tutorial on Meta-Network Models for Resilient Interdependent Infrastructures, Resilience Week, August 18-20, 2015, Philadelphia, PA, USA.
33. Organized Workshop on Game-Theoretic Approaches to Secure and Resilient Control Systems, American Control Conference (ACC), June 4-6, 2014, Portland, OR, USA.
34. Organized Workshop on Game-Theoretic Approaches to Secure and Resilient Control Systems, American Control Conference (ACC), June 17-19, 2013, Washington, DC, USA.

Scholarships, Awards and Honors

Fellowships and Scholarships

- Scholar in Residence, Digital Futures, KTH, 2022
- INFORMS Koopman Prize, 2022
- NYU Provost’s Global Research Initiatives (GRI) Fellowship, 2022
- NSF CAREER Award, 2019-2024
- NYU Goddard Fellowship, 2018
- NSERC Postdoctoral Fellowship, 2013-2014
- Ernest A. Reid Fellowship in Department of Electrical and Computer Engineering, UIUC, 2011-2012
- Mavis Future Faculty Fellow (MF3) Scholarship, College of Engineering, UIUC, 2010-2011
- Mavis Future Faculty Fellow (MF3) Scholarship, College of Engineering, UIUC, 2009-2010
- NSERC Canada Graduate Scholarship, 2007-2008
- University of Toronto ECE Fellowship, 2006-2007
- NSERC Undergraduate Summer Research Award, 2005-2006
- NSERC Undergraduate Industrial Research Award, 2005-2006

Best Paper Awards

- Best Paper Award: Y. Ge and Q. Zhu, “MEGA-PT: A Meta-game Framework for Agile Penetration Testing,” In: Proceedings of the International Conference on Decision and Game Theory for Security (GameSec 2024), pp. 24–44, 2024. https://doi.org/10.1007/978-3-031-74835-6_2.
- Best Student Paper Award: J. Huang and Q. Zhu, “PenHeal: A Two-Stage LLM Framework for Automated Pentesting and Optimal Remediation,” In: Proceedings of the Workshop on Autonomous Cybersecurity (AutonomousCyber ’24), ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 11–22, 2024. <https://doi.org/10.1145/3689933.3690831>.
- Best Paper Award: Y. Pan and Q. Zhu, “On Poisoned Wardrop Equilibrium in Congestion Games,” In: F. Fang, H. Xu, Y. Hayel (eds.), Decision and Game Theory for Security (GameSec 2022), Lecture Notes in Computer Science, vol. 13727. Springer, Cham, 2023. https://doi.org/10.1007/978-3-031-26369-9_10.
- Best Student Paper Award: L. Hang and Q. Zhu, “Combating Informational Denial-of-Service (IDoS) Attacks: Modeling and Mitigation of Attentional Human Vulnerability,” In: B. Bosansky, C. Gonzalez, S. Rass, A. Sinha (eds.), International Conference on Decision and Game Theory for Security (GameSec 2021), Lecture Notes in Computer Science, vol. 13061. Springer, Cham, 2021. https://doi.org/10.1007/978-3-030-90370-1_17.
- Finalist for Best Paper Award by a Young Scholar: Q. Zhu and T. Aneel, “Nash Equilibrium for Stochastic Differential-Algebraic Games with Applications in Control and Economy,” In: Proceedings of the 18th International Symposium on Dynamic Games and Applications, Grenoble, France, July 9-12, 2018.
- Winner of GlobalSIP 2017 Three-Minute Thesis (3MT) Competition: J. Pawlick, “Strategic and Physically-Aware Trust Management for the Internet of Controlled Things,” GlobalSIP 2017.
- Best Paper Award: W. Casey, Q. Zhu, J. Morales, and B. Mishra, “Compliance Control: Managed Vulnerability Surface in Socio-Technological Systems via Signaling Games,” In: Proceedings of the International Workshop on Managing Insider Security Threats (MIST 2015), 22nd ACM Conference on Computer and Communications Security (CCS), Denver, CO, October 12-16, 2015, pp. 53-62.
- Best Student Paper Runner-Up Award: R. Zhang † and Q. Zhu, “Secure and Resilient Distributed Machine Learning Under Adversarial Environments,” In: Proceedings of the International Conference on Information Fusion, Washington DC, July 6-9, 2015, pp. 644-651.
- Best Track Paper Award: Q. Zhu, C. Rieger, and T. Baar, “A Hierarchical Security Architecture for Cyber-Physical Systems,” In: Proceedings of the 4th International Symposium on Resilient Control Systems (ISRCs), Boise, ID, August 9-11, 2011.

Invited Talks

1. Keynote Speaker, LSU Symposium on Control, Learning, and Intelligent Systems, Louisiana State University, May 30, 2026.
2. Invited Speaker, “Coalition, Consensus, and Coordination: A Tale of Noncooperative and Cooperative Games in Socio-Technical Systems,” 12th Midwest Workshop on Control and Game Theory (MWCGT), Purdue University, West Lafayette, IN, April 25, 2026.
3. Invited Speaker, “The Role of Information in Dynamic Games and Multi-Agent Learning,” Industrial Engineering Seminar, University of Pittsburgh, Pittsburgh, PA, March 19, 2026.
4. Invited Visiting Professor, Hanyang University and National Seoul University, November 25-29, 2024.
5. Invited Speaker, 2024 INFORMS Analytics Conference, Orlando, FL, April 14-16, 2024.
6. Keynote Speaker, Georgia Tech Decision and Control Conference, February 23, 2024.
7. Invited Speaker, GE EDGE Symposium, September 19-21, 2024.
8. Invited Speaker, The Future of Cyber Deception ARO Workshop, CMU, Pittsburgh, PA, October 2-6, 2023.
9. Invited Speaker, Midwest Control and Game Theory Workshop, April 22-23, 2023.

10. Invited Participant, NSF Joint US-European Workshop on “Flexible Electric Grid Critical Infrastructure for Resilient Society,” Philadelphia, PA, April 21-22, 2023.
11. Invited Participant, Transatlantic US-Ireland-Northern Ireland Workshop on Collaborative IoT/CPS Cyber-security Research, October 20-21, 2022, CON- NECT Center, Dublin, Ireland.
12. Invited Participant, US-European Workshop on “Grid at the Edge: Towards the Zero-Carbon Power Grid with Improved Visibility, Safety, and Reliability,” Split, Croatia, May 23-24, 2022.
13. Invited Participant, NSF-Sponsored Workshop on “Grid at the Edge: Towards the Zero-Carbon Power Grid with Improved Visibility, Safety, and Reliability,” March 15 and March 22, 2022.
14. Invited Participant, “Facilitating U.S.-India Bilateral Research Collaborations Workshop,” NSF, March 15 and March 22, 2022.
15. Invited Participant, “URoL Workshop on Leveraging AI and Data Science for Predicting Mechanisms,” NSF, April 26, 2022.
16. Invited Participant, “Managing Industrial Control Systems Security Risks for Cyber Insurance,” Dagstuhl Seminars, Schloss Dagstuhl, Germany, November 7-12, 2021.
17. Invited Panelist, NSF S&CC PI Meeting Panel on COVID-19 Research, April 8, 2021 (Virtual).
18. Invited Participant, PSERC/NSF Workshop on “Grid at the Edge,” March 23-24, 2021 (Virtual).
19. Invited Participant, PSERC/NSF Executive Forum on “Grid at the Edge,” October 27-29, 2021 (Virtual).
20. Panelist, Symposium on the Science of Security (HotSoS), Discussion Groups on Adversaries, April 13-15, 2021 (Virtual).
21. Invited Speaker, “Game-Theoretic Assessment and Mitigation of Cyber Risks,” Risk Analytics Symposium 2021, March 26, 2021 (Virtual).
22. Invited Speaker, “Cross-Layer Cyber-Physical Security of Fission Battery Control Systems,” DOE Workshop on Safeguards and Security of Fission Batteries, April 2, 2021 (Virtual).
23. Invited Speaker, “Dynamic Games and Learning for Cyber Deception,” Dynamic Games and Applications Seminar, GERAD, Montreal, QC, December 3, 2020 (Virtual).
24. Keynote Speaker, “Dynamic Games and Learning for Cyber Deception,” AAAI- 20 Workshop on Artificial Intelligence for Cyber Security (AICS), February 8, 2020, New York City, NY.
25. Invited Speaker, “Supply Chain Risk and Mitigation for IoT-enabled Infrastructure Systems,” I3P Webinar, April 17, 2020.
26. Invited Speaker, “Toward a Game-Theoretic Foundation for Cyber Deception,” Summer 2019 Colloquium, Temple University, July 10, 2019.
27. Invited Participant, NSF/NAE Workshop on “Combating Climate Change and the ECE Renaissance,” University of Minnesota, April 19-20, 2019.
28. Invited Speaker, “Strategic Learning for Adaptive Defensive Deception,” ARO Workshop on Autonomous Adaptive Cyber Systems, April 23, 2019.
29. Invited Panelist, “Cyber Deception,” IEEE CNS, June 10-12, 2019, Washington, DC.
30. Invited Panelist, “Simulation for Cyber Risk Management,” Winter Simulation Conference, December 8-11, 2019, National Harbor, MD.
31. Keynote Speaker, 5th International Conference on Artificial Intelligence and Security (ICAIS), July 26-28, 2019, New York City, NY.
32. Invited Speaker, NSF Workshop on Energy Management Systems, Tokyo, Japan, June 20-21, 2019.
33. Invited Speaker, NSF Workshop on Energy Management Systems, IIT-Bombay, January 14-16, 2019.
34. Invited Lecturer, Mini-Course on Game-Theoretic Methods for Cyber-Physical Systems, University of Seville, October 22-26, 2018.

35. Distinguished Lecturer, ECE Distinguished Lecture Series, University of Connecticut, September 10, 2018.
36. Colloquium Speaker, "Dynamic Systemic Cyber Risk Control," Illinois Institute of Technology, October 1, 2018.
37. Invited Speaker, DHS CIRI Webinar, "Dynamic Resiliency Modeling and Planning," September 20, 2018.
38. Keynote Speaker, "Game Solving: Theory and Practice," 45th ICALP, July 9, 2018, Prague, Czech Republic.
39. Invited Panelist, DHS Cyber Risk Economics R&D Stakeholder Exchange Meeting, September 19, 2017, Arlington, VA.
40. Seminar Speaker, "Game Theory for Cyber-Physical Security," Tokyo Institute of Technology, June 21, 2017.
41. Seminar Speaker, "Resilient and Distributed Generation Control of Renewable Energies," JST-NSF Workshop, Tokyo, June 13, 2017.
42. Seminar Speaker, "Theoretic Foundation for Secure and Resilient IoBT," Army Research Lab, May 30, 2017.
43. Invited Speaker, "Game Theory for Cyber-Physical Security," CUNY Graduate Center, March 31, 2017.
44. Invited Speaker, "Resilient Analysis and Design of Interdependent Critical Infrastructures," Syracuse University, September 21-22, 2016.
45. Invited Speaker, "Game-Theoretic Framework for Microgrid Resiliency," MOPTA, Lehigh University, August 2016.
46. Invited Speaker, "Adversarial Machine Learning," Virginia Commonwealth University, September 2015.
47. Invited Speaker, "Resilient Control of Complex Networks," IBM Watson Research, December 2014.
48. Invited Participant, NSF Ideas Lab on Secure Cyber-Physical Systems, August 2014.
49. Invited Speaker, "Game Theory: From Economics to Engineering," CSpash, Courant Institute, April 2014.
50. Seminar Speaker, Secure and Resilient Dynamic Routing in Cognitive Radio Networks, Department of Electrical and Computer Engineering, University of Delaware, April 28, 2014.

Services

Departmental Services

- Served on 2024 hiring committee.
- Served on 2021 hiring committee to determine the cluster area to hire in the next five years.
- Arranged and planned departmental seminars, in charge of ECE-GY 9900 Seminar in Electrical & Computer Engineering.
- Planned and organized admitted PhD student day on March 8, 2018.
- Initiated the first Vertically Integrated Project (VIP) of the ECE department.
- Supervised a team of VIP undergraduate student project.
- Served on 2017 retreat committee.
- Served on 2018 retreat subcommittee on graduate recruitment.
- Co-organized Ph.D. Admitted Student Day and ECE Open House on March 08, 2018.
- Supervised a total 22 NYU undergraduate students for summer undergraduate research program: – 3 summer undergraduate students in 2015 – 5 summer undergraduate students in 2016 – 4 summer undergraduate students in 2017 – 3 summer undergraduate students in 2018 – 1 summer undergraduate students in 2019 – 1 summer undergraduate students in 2020 – 2 summer undergraduate students in 2021 – 1 summer undergraduate students in 2022 – 1 summer undergraduate students in 2023 – 1 summer undergraduate students in 2024

School Services

- Initiated Tandon-wide activities to prevent and respond to faculty Bullying and Other Demeaning and disruptive behavior (BORDER).
- Served as Editor-in-Chief (EiC) of Cyber Byte Newsletter of Center of Cyber Security (CCS), April 2021-Present.
- Faculty Engineers in Residence (F-EIR) to bridge the gap between academia and entrepreneurship.
- Spearheaded the first New York City Multidisciplinary Symposium on Security and Privacy, Brooklyn, NY, held on Feb. 25, 2017.
- Invited to serve on PhD Dissertation Committee of
- Juan Herrera, PhD student from NYU Steinhardt
- Wilmer Cantos, PhD student from Department of Civil and Urban Engineering, NYU
- Karel Durkota, PhD Student from Czech Technical University in Prague, Czech Republic
- Sharad Kumar Singh, PhD Student from India Institute of Technology, Madras, India
- M.Venkateswararao.K, PhD Student from India Institute of Technology, Bombay, India
- Arnold Kouam Kounchou, PhD Student from University of Avignon, France
- Bo Pang, PhD Student at ECE NYU
- Leilei Cui, PhD Student at ECE NYU
- Mengzhe Huang, PhD Student at ECE NYU

University Services

- Served as the NYU representative of Institute for Information Infrastructure Protection (I3P).
- Faculty advisor for newly established NYU International Council on Systems Engineering (INCOSE) Student Chapter:<https://wp.nyu.edu/nyuincose/>.
- Developing Cooperative Research And Development Agreement (CRADA) between NYU and Army Research Laboratory (ARL).
- Initiated NYU collaboration with University of Strathclyde, Glasgow, UK on power systems research.

Editorship

- Guest Editor, IEEE Transactions on Industrial Cyber-Physical Systems Special Collection on Co-design of Cognition, Control, Computing and Communication for Industrial Cyber-Physical Systems, 2026.
- Guest Editor, IEEE Open Journal of the Communications Society Special issue on Trustworthy AI-Enabled Edge Computing in Next-Generation Wireless Networks, 2024
- Guest Editor, IEEE Journal on Selected Areas in Communications (JSAC) Special Issue on Co-Design of Communication, Computing, and Control in Industrial Cyber-Physical Systems, 2024.
- Guest Editor, IEEE Network Magazine Special Issue on Network-native Security, Safety, and Resilience Solutions for Critical Automation Systems Hosted in Cloud and Fog Computing, 2024.
- Guest Editor, IEEE Transactions on Control Systems Technology (TCST) Special Issue on Resilient Control of Cyber Physical Energy Systems, 2023.
- Guest Editor, Special Issue on Dynamic Games for Control of Epidemics, Dynamic Games and Applications (DGAA), Elsevier, 2021.
- Guest Editor, Special Section on Learning and Security in Multi-Agent Systems, Annual Review in Control (ARC), Elsevier, 2021.

- Associate Editor, IEEE Transactions on Aerospace and Electronic Systems, 2022- Present.
- Associate Editor, IEEE Transactions on Network Science and Engineering, 2024- Present.
- Associate Editor, IEEE Transactions on Network and Service Management (TNSM), Present.
- Associate Editor, Dynamic Games and Applications (DGAA), Elsevier, 2021- Present.
- Associate Editor in the area of Management of Energy Demand, Storage, and EVs, IET Transaction on Smart Grid, 2019-Present.
- Guest Editor of Special Issue on Game Theory for Cyber Security, Computers & Security, Elsevier, 2018
- Guest Editor, MDPI Special Issues, Collections and Topics in game theory, cybersecurity, resilient cyber-physical-human systems, AI methods in automation, Internet of Things, critical infrastructures, stochastic control and games, decentralized control, and human factors in control systems.
- Guest Editor of Special Issue on Game Models for Cyber-Physical Infrastructures, Games, MDPI, 2018.

Scientific Committees and Activities

- Program Committee Member, Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES 2026), Malmö, Sweden, October 12–14, 2026.
- Member, Executive Board, The International Society of Dynamic Games (ISDG).
- Member, International Program Committee, 21st International Symposium on Dynamic Games and Applications (ISDG), July 20–23, 2026.
- TPC Member, 13th International Conference on Networks, Games, Control and Optimization (NETGCOOP), Palermo, Italy, October 14–16, 2026.
- Initiated NYU–TAU Cyber Collaboration on Human-Centric Cyber Resilience, May 2025.
- Board Member, INFORMS Military and Security Society (MAS), 2025.
- Chair, 15th Conference on Decision and Game Theory for Security (GameSec), New York, NY, October 16-18, 2024.
- TPC Member, Tech4Good: Economic and Computational Advances in Distributed Systems, Co-located with ICDCS 2024, Jersey City, NJ, June 16, 2024.
- TPC Member, 23rd Workshop on the Economics of Information Security, Dallas, TX, April 8-10, 2024.
- Organizer, ARO Workshop on Interdisciplinary Cyber Deception, New York, NY, August 17-18, 2023.
- Advisor, Inaugural NYC Intelligent Transportation System Workshop: Network Designs in the Era of Digital Mobility, NYU, January 16, 2024.
- Advisor, Inaugural NYC Cybersecurity Day Workshop: Human Elements in Cybersecurity, NYU, November 20, 2023.
- Member, Advisory Committee, Game Theory for Cyber Security (GameSec), Conference on Decision and Game Theory for Security, October 25-27, 2021, Prague, Czech Republic (Virtual).
- Founding Chair, Technical Committee on Security and Privacy, IEEE Control Systems Society, 2021-Present.
- TPC Chair, Resilient Control Systems, Resilience Week 2020, Salt Lake City, UT, October 19-22, 2020 (Virtual).
- TPC Chair, Resilient Control Systems, Resilience Week 2021, Washington, DC, October 18-21, 2021 (Virtual).
- Theme Chair, International Symposium on Sustainable Systems and Technologies (ISSST) 2020 on Critical Interdependent Infrastructure Systems, rescheduled to Pittsburgh, June 22-24, 2021.
- TPC Chair, 11th Conference on Decision and Game Theory for Security (GameSec), October 28-30, 2020, College Park, MD, USA.
- TPC Chair, IEEE International Workshop on Information Forensics and Security (WIFS), December 6-9, 2020, New York City, NY, USA.

- Vice-General Chair, International Conference on NETwork Games, COntrol and OPTimisation (NETGCOOP), New York, NY, November 14-16, 2018.
- General Chair, 5th International Conference on Artificial Intelligence and Security (ICAIS), July 26-28, 2019, New York University, New York, NY, USA.
- Organizer, 1st New York Multidisciplinary Symposium on Security and Privacy, Spring 2017.
- General Chair, 7th Conference on Decision and Game Theory for Security (GameSec), November 2-4, 2016, New York, NY, USA.
- Member, Organizing Committee, NSF National Workshop on Resilience Research (NWRR) for Critical Infrastructure, Arlington, VA, October 22-23, 2016.
- Advisory Panel Member, Engineering Higher Education, collaborated with CollegeBoard to develop curriculum outlines for the Advanced Placement (AP) Program in Engineering.
- Initiator, 1st US-European Workshop on Interdependent Infrastructure Networks (WI2N), January 15-16, 2015, New York University.
- Organizer, 2015 NYU Workshop on Control and Optimization of Network Systems (CONES), May 1, 2015.
- Member, Technical Program Committees:
- IEEE International Conference on Computer Communications (INFOCOM), 2026.
- IEEE Global Communications Conference (GLOBECOM), Selected Areas in Communications Symposium: Smart Grid Communications and Control, Macau, China, December 7–11, 2026.
- Conference on New Innovations in Material Science (NIMS), 2026.
- International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), 2026.
- International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt), 2026.
- IEEE International Conference on Computer Communications (INFOCOM), 2025.
- IEEE/IFIP Network Operations and Management Symposium (NOMS), 2025.
- International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), 2025.
- Symposium and Bootcamp on the Science of Security (HotSoS), 2018.
- ACM Workshop on Cyber-Physical Systems Security and Privacy (CPS-SPC), 2015–2018.
- IEEE International Conference on Smart Computing (SMARTCOMP), 2017.
- IFIP WG 11.3 Conference on Data and Applications Security and Privacy (DBSec), 2017.
- IEEE Conference on Communications and Network Security (CNS), 2016.
- IEEE Real-Time Systems Symposium (RTSS), 2016.
- Conference on Decision and Game Theory for Security (GameSec), 2015–2024.
- International Conference on Performance Evaluation Methodologies and Tools (VALUETOOLS), 2015–2024.
- International Conference on Network Games, Control and Optimization (NETGCOOP), 2015, 2016.
- International Conference on Game Theory for Networks (GameNets), 2014.
- International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt), 2015–2024.
- ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), 2015.
- IEEE Global Communications Conference, Selected Areas in Communications Symposium: Smart Grid Communications, 2015.

NSF Panel Services

- Served on NSF ECCS EPCL Panel, 2026.
- Served on NSF CPS-FR Panel, 2026.
- Served on NSF CPS CAREER Panel, 2025.
- Served on NSF CMMI DCSD Panel, 2024.
- Served on NSF CPS Medium Panel, 2023.
- Served on NSF LEAP-HI Panel, 2022.
- Served on NSF EPCN CAREER Panel, 2022.
- Served on NSF CPS CAREER Panel, 2022.
- Served on NSF Smart and Connected Health (SCH) Review Panel, 2021.
- Served on NSF CNS Core Program Review Panel, 2021.
- Served on NSF SaTC Review Panel, 2020, 2021.
- Served on NSF Smart and Connected Community (SCC) Review Panel, 2020.
- Served on NSF Graduate Research Fellowship Program (GRFP) Review Panel, 2020.
- Served on NSF CPS Review Panel, 2019.
- Served on NSF NeTS Review Panel, 2019.
- Served on NSF CPS Review Panel, 2018.
- Served on NSF CSSI Review Panel, 2018.
- Served on NSF SaTC Review Panel, 2017.
- Served on NSF CPS Review Panel, 2016.
- Served on NSF CAREER Panel, 2016.
- Served on NSF EPAS Panel, 2014, and EPCN Panel, 2015, 2016.
- Served on NSF ERC Review Panel on Energy and Climate, 2015.